

Citations in the Slides

Incentivizing Retail Traders (Presentation V4)

Every citation shown on the slides (29 papers), in the order they first appear in the talk. Each entry gives how it is labelled on the slide, where it appears and the bullet it supports, then the reference, links, what the draft says, and the abstract, taken from `cites.docx`. Two papers added for the slides (Ray 2009; Anginer et al. 2024) are not in `cites.docx`; their details come from Crossref and OpenAlex. Corrections flagged in `cites.docx` are repeated under the relevant entry.

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1. D’Acunto et al. 2019b

cites.docx entry 18 · key: rossi2019

Slide 2, Technology is Changing How Retail Investors Trade: “Spending (D’Acunto et al. 2019b) and saving (Gargano and Rossi 2024)”

Crowdsourcing Financial Information to Change Spending Behavior. Working paper, SSRN 3348722 (CE-Sifo Working Paper No. 7533). Since published under a new title: “Crowdsourcing peer information to change spending behavior,” Journal of Financial Economics, 157 (2024), article 103858.

Authors: Francesco D’Acunto, Alberto G. Rossi, Michael Weber

SSRN: papers.ssrn.com/abstract_id=3348722 · [DOI of the JFE version](#)

Claims in the draft:

- p. 1, introduction: Fintech innovations have been shown to influence spending.
- p. 4, contribution: Fintech can reduce overspending.

Abstract (SSRN working paper, the version the draft cites):

We document five effects of providing individuals with crowdsourced spending information about their peers (individuals with similar characteristics) through a FinTech app. First, users who spend more than their peers reduce their spending significantly, whereas users who spend less keep constant or increase their spending. Second, users’ distance from their peers’ spending affects the reaction monotonically in both directions. Third, users’ reaction is asymmetric - spending cuts are three times as large as increases. Fourth, lower-income users react more than others. Fifth, discretionary spending drives the reaction in both directions and especially cash withdrawals, which are commonly used for incidental expenses and anonymous transactions. We argue Bayesian updating, peer pressure, or the fact that bad news looms more than (equally-sized) good news cannot alone explain all these facts.

Abstract (published JFE version):

We isolate the information channel of peer effects in consumption in a setting that excludes a role for common shocks or social pressure—a spending panel paired with crowdsourced information about anonymous “peers” elicited at different times. Consumers converge to peers’ spending, and more so when peer signals are more informative. Convergence is asymmetric: within 12 months of information provision, overspenders close 17% and underspenders 5% of their gap relative to peers. We exploit the quasi-random assignment to peer groups in an instrumental-variable strategy and implement an experiment for external validity. Our results are consistent with information-based theories of overconsumption.

Correction flagged in cites.docx: draft says “Albtero”; SSRN working paper; published record: Alberto; now “Crowdsourcing peer information to change spending behavior,” JFE 157 (2024), 103858

2. Gargano and Rossi 2024

cites.docx entry 28 · key: gargano2024

Slide 2, Technology is Changing How Retail Investors Trade: “Spending (D’Acunto et al. 2019b) and saving (Gargano and Rossi 2024)”

Slide 7, Contribution: “Fintech platform design can help (D’Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)”

Goal Setting and Saving in the FinTech Era. The Journal of Finance, 79(3), 1931–1976.

Authors: Antonio Gargano, Alberto G. Rossi

SSRN: papers.ssrn.com/abstract_id=3579275 · DOI

Claims in the draft:

- p. 1, introduction: Fintech innovations have been shown to influence saving behavior.
- p. 4, contribution: Fintech can increase savings.

Abstract (published version):

We study the effectiveness of saving goals in increasing individuals' savings using data from a Fintech app. Using a difference-in-differences identification strategy that randomly assigns users into a group of beta testers who can set goals and a group of users who cannot, we find that setting goals increases individuals' savings rate. The increased savings within the app do not reduce savings outside the app. Moreover, goal setting helps those individuals previously identified as having the lowest propensity to save. Matching App user survey responses to their behavior highlights the relative merits of monitoring and concreteness channels in explaining our findings.

3. D'Acunto et al. 2019a

cites.docx entry 17 · key: acunto

Slide 2, Technology is Changing How Retail Investors Trade: “Robo-advisors help investors diversify (**D'Acunto et al. 2019a**)”

Slide 7, Contribution: “Fintech platform design can help (**D'Acunto et al. 2019a**; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)”

The Promises and Pitfalls of Robo-Advising. *The Review of Financial Studies*, 32(5), 1983–2020.

Authors: Francesco D'Acunto, Nagpurnanand Prabhala, Alberto G. Rossi

SSRN: papers.ssrn.com/abstract_id=3122577 (authors' posting) · [3165339](https://www.cesifo.org/en/papers/working/3165339) (CESifo working-paper record) · DOI

Claims in the draft:

- p. 1, introduction: Robo-advisors help investors diversify.
- p. 3, contribution (with Rossi and Utkus 2021): Fintech can help households improve their trading performance and asset allocation.

Abstract (published version):

We study the introduction of a wealth-management robo-adviser that constructs portfolios tailored to investors' holdings and preferences. Adopters are similar to non-adopters in terms of demographics and prior interactions with human advisers but tend to be more active and have greater assets under management. Investors adopting robo-advising experience diversification benefits. Ex ante undiversified investors increase stock holdings and hold portfolios with less volatility and better returns. Already well-diversified investors hold fewer stocks, yet see some reduction in volatility, and trade more after adoption. All investors increase attention based on online account logins. We find that adopters exhibit declines in prominent behavioral biases, including the disposition, trend chasing, and rank effect. Our results emphasize the promises and pitfalls of robo-advising tools, which are becoming ubiquitous all over the world.

4. Barber et al. 2022

cites.docx entry 4 · key: schwarz_2022

Slide 2, Technology is Changing How Retail Investors Trade: “‘Gamified’ interfaces alter investor attention (Barber et al. 2022; Tan 2021)”

Slide 7, Contribution: “Fintech platform design can help (D’Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)”

Attention-Induced Trading and Returns: Evidence from Robinhood Users. *The Journal of Finance*, 77(6), 3141–3190.

Authors: Brad M. Barber, Xing Huang, Terrance Odean, Christopher Schwarz

SSRN: papers.ssrn.com/abstract_id=3715077 · [DOI](#)

Claims in the draft:

- p. 1, introduction (with Tan 2021): The retail brokerage Robinhood introduced “gamified” interfaces that alter investor attention.
- p. 4, contribution: Fintech innovations can hurt users by focusing trading on high-attention stocks.

Abstract (published version):

We study the influence of financial innovation by fintech brokerages on individual investors’ trading and stock prices. Using data from Robinhood, we find that Robinhood investors engage in more attention-induced trading than other retail investors. For example, Robinhood outages disproportionately reduce trading in high-attention stocks. While this evidence is consistent with Robinhood attracting relatively inexperienced investors, we show that it is also driven in part by the app’s unique features. Consistent with models of attention-induced trading, intense buying by Robinhood users forecasts negative returns. Average 20-day abnormal returns are −4.7% for the top stocks purchased each day.

5. Tan 2021

cites.docx entry 52 · key: tan2021

Slide 2, Technology is Changing How Retail Investors Trade: “‘Gamified’ interfaces alter investor attention (Barber et al. 2022; Tan 2021)”

Slide 7, Contribution: “Fintech platform design can help (D’Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)”

Democratizing Finance with Robinhood: Financial Infrastructure, Interface Design and Platform Capitalism. *Environment and Planning A: Economy and Space*, 53(8), 1862–1878.

Authors: Gordon Kuo Siong Tan

SSRN: No SSRN version found. [DOI](#)

Claims in the draft:

- p. 1, introduction (with Barber et al. 2022): The retail brokerage Robinhood introduced “gamified” interfaces that alter investor attention.
- p. 4, contribution: Fintech innovations can hurt users by incentivizing greater risk taking.

Abstract:

It is widely claimed that financial technology democratizes financial access and promotes financial inclusion. This paper challenges this dominant narrative of financial technology using the popular US-based

mobile investing platform Robinhood as a case study. Analyzing press articles on Robinhood and regulatory filings of online brokerages, the concepts of financial infrastructures and platforms are used to unpack the capitalist logics that drive the platform business model. Specifically, this paper shows how digital interface design plays a central role in Robinhood's articulation of investors into the stock markets, through its simple, minimalist app that keeps users engaged by the productive management of 'frictions'. The ease of investing via platform-based brokerages has seen investors taking on greater amounts of risk. This paper argues that Robinhood's success is driven by the continued expansion of its user base using various interface design techniques. This allows platform capitalism to be enacted by extracting rent through various revenue streams, where higher rents are derived from more frequent and riskier trading behaviors. The narrative of democratizing finance as employed widely in the financial technology sector thus obscures the capitalist logics and predatory practices that underlie financial technology.

6. Ma et al. 2019

cites.docx entry 40 · key: ma2019

Slide 3, Social Trading: Retail Traders as Fund Managers: "An institutional high water mark contract, but paid daily to retail traders (Ma et al. 2019; Aragon and Nanda 2012)"

Portfolio Manager Compensation in the U.S. Mutual Fund Industry. *The Journal of Finance*, 74(2), 587–638.

Authors: Linlin Ma, Yuehua Tang, Juan-Pedro Gómez

SSRN: papers.ssrn.com/abstract_id=2024027 · DOI

Claim in the draft (p. 1, introduction): Ma et al. find that 79% of mutual funds in their sample use performance-based pay.

Abstract:

We study compensation contracts of individual portfolio managers using hand-collected data of over 4,500 U.S. mutual funds. Variations in the compensation structures are broadly consistent with an optimal contracting equilibrium. The likelihood of explicit performance-based incentives is positively correlated with the intensity of agency conflicts, as proxied by the advisor's clientele dispersion, its affiliations in the financial industry, and its ownership structure. Investor sophistication and the threat of dismissal in outsourced funds serve as substitutes for explicit performance-based incentives. Finally, we find little evidence of differences in future performance associated with any particular compensation arrangement.

Note: The abstract gives no percentage. The authors' own summary on the [Harvard Law School Forum on Corporate Governance](#) says: "the bonus component of compensation is explicitly tied to the fund's investment performance for 79.0% of sample funds." That is the authors' summary post, not the journal text, which was not accessed.

7. Aragon and Nanda 2012

cites.docx entry 3 · key: aragon2012

Slide 3, Social Trading: Retail Traders as Fund Managers: "An institutional high water mark contract, but paid daily to retail traders (Ma et al. 2019; Aragon and Nanda 2012)"

Tournament Behavior in Hedge Funds: High-water Marks, Fund Liquidation, and Managerial Stake. *The Review of Financial Studies*, 25(3), 937–974.

Authors: George O. Aragon, Vikram Nanda

SSRN: papers.ssrn.com/abstract_id=1570789 · DOI. The SSRN listing carries the earlier working-paper title, "On Tournament Behavior in Hedge Funds: High Water Marks, Managerial Horizon, and the Backfilling Bias"; the authors match, so this is a likely rather than confirmed match.

Claim in the draft (p. 1, introduction): Aragon and Nanda discuss how hedge funds commonly use this asymmetric incentive structure (high water marks) in conjunction with long-horizon return benchmarks.

Abstract (published version):

We analyze whether risk shifting by a hedge fund manager is related to the manager's incentive contract, personal capital stake, and the risk of fund closure. We find that the propensity to increase risk following poor performance is significantly weaker when incentive pay is tied to the fund's high-water mark and when funds face little immediate risk of liquidation. Risk shifting is also less prevalent when a manager has a significant amount of personal capital invested in the fund. Overall, high-water mark provisions, managerial stake, and low risk of fund closure appear to make a hedge fund manager more conservative with regard to risk shifting.

Correction flagged in cites.docx: draft says 2011; published record: 2012 for RFS 25(3); 2011 is the advance-access year

8. Jensen and Meckling 1976

cites.docx entry 36 · key: jensen_meckling

Slide 4, How Could Daily Incentives Affect Retail Traders?: "Performance pay aligns trader and investor interests, potentially mitigating retail biases (**Jensen and Meckling 1976**; Holmstrom 1979)"

Slide 7, Contribution: "Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (**Jensen and Meckling 1976**; Holmstrom 1979)"

Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure. *Journal of Financial Economics*, 3(4), 305–360.

Authors: Michael C. Jensen, William H. Meckling

SSRN: papers.ssrn.com/abstract=94043 · DOI

Claim in the draft (p. 2, introduction): Cited with Holmstrom (1979) for the statement that principal-agent theory suggests performance pay can align interests.

Abstract (published version):

This paper integrates elements from the theory of agency, the theory of property rights and the theory of finance to develop a theory of the ownership structure of the firm. We define the concept of agency costs, show its relationship to the 'separation and control' issue, investigate the nature of the agency costs generated by the existence of debt and outside equity, demonstrate who bears these costs and why, and investigate the Pareto optimality of their existence. We also provide a new definition of the firm, and show how our analysis of the factors influencing the creation and issuance of debt and equity claims is a special case of the supply side of the completeness of markets problem.

9. Holmstrom 1979

cites.docx entry 35 · key: holmstrom

Slide 4, How Could Daily Incentives Affect Retail Traders?: “Performance pay aligns trader and investor interests, potentially mitigating retail biases (Jensen and Meckling 1976; **Holmstrom 1979**)”

Slide 7, Contribution: “Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; **Holmstrom 1979**)”

Moral Hazard and Observability. The Bell Journal of Economics, 10(1), 74–91.

Authors: Bengt Holmstrom (Holmström)

SSRN: No SSRN version found. [DOI / JSTOR](#)

Claim in the draft (p. 2, introduction): Cited with Jensen and Meckling (1976) for the statement that principal-agent theory suggests performance pay can align interests.

Abstract:

The role of imperfect information in a principal-agent relationship subject to moral hazard is considered. A necessary and sufficient condition for imperfect information to improve on contracts based on the payoff alone is derived, and a characterization of the optimal use of such information is given.

10. Odean 1998

cites.docx entry 42 · key: odean1998_disposition

Slide 4, How Could Daily Incentives Affect Retail Traders?: “Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking (**Odean 1998**; Odean 1999; Dorn et al. 2015; Delfabbro et al. 2021)”

Are Investors Reluctant to Realize Their Losses? The Journal of Finance, 53(5), 1775–1798.

Authors: Terrance Odean

SSRN: papers.ssrn.com/abstract_id=94142 · [DOI](#)

Claim in the draft (p. 2, introduction): The extreme frequency of the incentives might amplify detrimental behaviors like loss aversion.

Abstract (SSRN working-paper version):

I test the disposition effect, the tendency of investors to hold losing investments too long and sell winning investments too soon, by analyzing trading records for 10,000 accounts at a large discount brokerage house. These investors demonstrate a strong preference for realizing winners rather than losers. Their behavior does not appear to be motivated by a desire to rebalance portfolios, or to avoid the higher trading costs of low price stocks. Nor is it justified by subsequent portfolio performance. For taxable investments, it is sub-optimal and leads to lower after-tax returns. Tax-motivated selling is most evident in December.

11. Odean 1999

cites.docx entry 43 · key: odean1999

Slide 4, How Could Daily Incentives Affect Retail Traders?: “Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking (Odean 1998; **Odean 1999**; Dorn et al. 2015; Delfabbro et al. 2021)”

Do Investors Trade Too Much? American Economic Review, 89(5), 1279–1298.

Authors: Terrance Odean

SSRN: papers.ssrn.com/abstract_id=94143 · second record [219248](#) · DOI

Claims in the draft (both p. 2, introduction):

- With Barber and Odean (2008): Retail traders are attention driven and overconfident.
- The extreme frequency of the incentives might amplify excessive trading.

Abstract (SSRN working-paper version; no abstract was found for the published AER article):

This paper takes a first step towards demonstrating that overall trading volume in equity markets is excessive, by showing that it is excessive for a particular group of investors: those with discount brokerage accounts. One possible cause of excessive trading is overconfidence. Overconfident investors will trade too frequently, that is, the gains overconfident investors realize through trade will be less than they anticipate and may not even offset the costs of trading. By analyzing trading records for 10,000 accounts at a large discount brokerage house, I test whether the securities these investors purchase outperform those they sell by enough to cover the costs of trading. I find the surprising result that, on average, the securities they purchase actually underperform those they sell. This is the case even when trading is not apparently motivated by liquidity demands, tax-loss selling, portfolio rebalancing, or a move to lower-risk securities. I examine return patterns before and after transactions. Return patterns before purchases and sales can be explained by the difficulty of the search for securities to buy, investors' tendency to let their attention be directed by outside sources, the disposition effect, and investors' reluctance to sell short.

12. Dorn et al. 2015

cites.docx entry 22 · key: dorn2014

Slide 4, How Could Daily Incentives Affect Retail Traders?: “Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking (Odean 1998; Odean 1999; **Dorn et al. 2015**; Delfabbro et al. 2021)”

Trading as Gambling. *Management Science*, 61(10), 2376–2393.

Authors: Anne Jones Dorn, Daniel Dorn, Paul Sengmueller

SSRN: papers.ssrn.com/abstract_id=2160996 · DOI. The SSRN record is a 2012 working paper whose abstract describes two samples; the published abstract below describes three.

Claim in the draft (p. 2, introduction): Cited with Delfabbro et al. (2021) for the statement that the extreme frequency of the incentives might amplify short-term thinking.

Abstract (published version):

This paper offers evidence from three different samples consistent with investors substituting between playing the lottery and gambling in financial markets. In the United States, increases in the jackpots of the multistate lotteries Powerball and Mega Millions are associated with significant reductions in small trade participation in the stock market. California-based discount brokerage clients and German discount brokerage clients are significantly less likely to trade during weeks with larger lottery prizes in the California and German lotteries, respectively. Variation in lottery prizes affects speculative trading in more lottery-like securities such as individual stocks and options, but not trading in bonds and mutual funds. Trading that is likely associated with long-term savings motives, such as trading in retirement accounts, does not respond to lottery jackpots, either. The negative relation between trading activity and jackpots is stronger for individuals who are more likely to play the lottery.

Correction flagged in cites.docx: draft says 2014; published record: 2015 for *Management Science* 61(10)

13. Delfabbro et al. 2021

cites.docx entry 19 · key: delfabbro2021

Slide 4, How Could Daily Incentives Affect Retail Traders?: “Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking (Odean 1998; Odean 1999; Dorn et al. 2015; **Delfabbro et al. 2021**)”

Cryptocurrency Trading, Gambling and Problem Gambling. Addictive Behaviors, 122, article 107021.

Authors: Paul Delfabbro, Daniel King, Jennifer Williams, Neophytos Georgiou

SSRN: No SSRN version found. [DOI](#)

Claim in the draft (p. 2, introduction): Cited with Dorn et al. for the statement that the extreme frequency of the incentives might amplify short-term thinking.

Abstract:

Some forms of speculative trading share similarities with gambling. Decisions are often based on limited information, short-term motives for gain, and highly volatile and uncertain outcomes. Given these similarities, there is evidence to show that people who are attracted to gambling are also statistically more likely to engage in higher risk speculation such as day-trading of stocks and crypto-currency trading. In this study, involving 543 people (M = 388, F = 155, 85% aged 18–40 years) who reported at least monthly sports-betting, crypto-currency trading or both, we examined whether gambling and problem gambling were reliable predictors of the reported intensity of crypto-currency trading. The results showed that gambling and problem gambling rates were highest among those who reported both activities and that problem gambling scores (PGSI) and engaging in stock trading was significantly related to measures of crypto-currency trading intensity as based on the time spent per day, number of trades and level of expenditure. Future research should examine whether gambling history and involvement influences how people manage their investments in crypto-currencies, including their propensity for making riskier decisions and experiencing more negative outcomes.

14. de Chaisemartin and D’Haultfoeuille 2026

cites.docx entry 15 · key: clement_xavier

Slide 5, Data and Methodology: “Differences-in-differences around each bonus (**de Chaisemartin and D’Haultfoeuille 2026**)”

Slide 32, Differences-in-Differences Estimator: “Estimator of **de Chaisemartin and D’Haultfoeuille 2026**: change from the day before a bonus, vs. funds with the same treatment history not yet treated”

Difference-in-Differences Estimators of Intertemporal Treatment Effects. The Review of Economics and Statistics, 108(4), 863–880 (2026, per Crossref). The draft gives 2024 and no volume or pages.

Authors: Clément de Chaisemartin, Xavier D’Haultfoeuille

SSRN: papers.ssrn.com/abstract_id=3731856 · [DOI](#) · [arXiv](#) · [NBER w29873](#)

Claims in the draft:

- p. 3, introduction: The draft employs this paper’s differences-in-differences design, which can handle the repeated and non-absorbing treatments (bonuses) in this setting.
- p. 18, Section 4: The estimator adapts DiD techniques to estimate the effects of repeated and non-absorbing treatments. The draft describes the methodology as defining treatment as a cumulative measure of past bonus achievements and estimating a dynamic treatment effect model, with time-relative indicators measured from the last period a trader was untreated before earning a new bonus. Equation (4) is presented as a simplified form of this design.

Abstract (published version):

We study treatment-effect estimation using panel data. The treatment may be nonbinary, nonabsorbing, and the outcome may be affected by treatment lags. We make a parallel-trends assumption and propose event-study estimators of the effect of being exposed to a weakly higher treatment dose for ℓ periods. We also propose normalized estimators that estimate a weighted average of the effects of the current treatment and its lags. We also analyze commonly used two-way fixed-effects regressions. Unlike our estimators, they can be biased in the presence of heterogeneous treatment effects. A local-projection version of those regressions is biased even with homogeneous effects.

Correction flagged in cites.docx: draft says 2024, no volume or pages; published record: REStat 108(4), 863–880, 2026 per Crossref; 2024 is probably the online-early year (inference)

15. Brown et al. 1996

cites.docx entry 11 · key: brown1996

Slide 7, Contribution: “Performance incentives are well studied for institutional managers (**Brown et al. 1996**; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)”

Of Tournaments and Temptations: An Analysis of Managerial Incentives in the Mutual Fund Industry. The Journal of Finance, 51(1), 85–110.

Authors: Keith C. Brown, W. V. Harlow, Laura T. Starks

SSRN: papers.ssrn.com/abstract_id=7460 · DOI

Claim in the draft (p. 2, introduction): When funds underperform relative to their peers, they tend to increase volatility in an effort to catch up.

Abstract (published version):

We test the hypothesis that when their compensation is linked to relative performance, managers of investment portfolios likely to end up as “losers” will manipulate fund risk differently than those managing portfolios likely to be “winners.” An empirical investigation of the performance of 334 growth-oriented mutual funds during 1976 to 1991 demonstrates that mid-year losers tend to increase fund volatility in the latter part of an annual assessment period to a greater extent than mid-year winners. Furthermore, we show that this effect became stronger as industry growth and investor awareness of fund performance increased over time.

16. Massa and Patgiri 2009

cites.docx entry 41 · key: massa2009

Slide 7, Contribution: “Performance incentives are well studied for institutional managers (Brown et al. 1996; **Massa and Patgiri 2009**; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)”

Incentives and Mutual Fund Performance: Higher Performance or Just Higher Risk Taking? The Review of Financial Studies, 22(5), 1777–1815.

Authors: Massimo Massa, Rajdeep Patgiri

SSRN: papers.ssrn.com/abstract_id=1376201 · DOI

Claim in the draft (p. 2, introduction): Cited with Buraschi et al. (2014) as showing that high water mark incentives induce risk taking and motivate managers to use more leverage.

Abstract:

We study the impact of contractual incentives on the performance of mutual funds. We find that high-incentive contracts induce managers to take more risk and reduce the funds' probability of survival. Yet, funds with high-incentive contracts deliver higher risk-adjusted return, and the superior performance remains persistent. The top incentive quintile of funds outperforms the bottom quintile by 2.70% per year. Moreover, high-incentive winner funds from one year have a positive alpha of 0.41% per month in the following year. Focusing on funds' holdings, we show that active portfolio rebalancing is the main channel through which incentives increase performance.

Note (my reading of the abstract only): The abstract is about contractual incentives in mutual funds and supports the risk-taking part of the claim. It does not mention high water marks or leverage.

Correction flagged in cites.docx: draft says 2005; published record: 2009 for RFS 22(5)

17. Agarwal et al. 2009

cites.docx entry 1 · key: agarwal2009

Slide 7, Contribution: “Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; **Agarwal et al. 2009**; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)”

Role of Managerial Incentives and Discretion in Hedge Fund Performance. The Journal of Finance, 64(5), 2221–2256.

Authors: Vikas Agarwal, Naveen D. Daniel, Narayan Y. Naik

SSRN: papers.ssrn.com/abstract_id=889008 · DOI

Claim in the draft (p. 2, introduction): The higher risk levels brought on by high water marks correspond with greater risk-adjusted returns among hedge funds.

Abstract:

Using a comprehensive hedge fund database, we examine the role of managerial incentives and discretion in hedge fund performance. Hedge funds with greater managerial incentives, proxied by the delta of the option-like incentive fee contracts, higher levels of managerial ownership, and the inclusion of high-water mark provisions in the incentive contracts, are associated with superior performance. The incentive fee percentage rate by itself does not explain performance. We also find that funds with a higher degree of managerial discretion, proxied by longer lockup, notice, and redemption periods, deliver superior performance. These results are robust to using alternative performance measures and controlling for different data-related biases.

18. Ray 2009

not in cites.docx · key: ray2009

Slide 7, Contribution: “Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; **Ray 2009**); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)”

The Downside of High Water Marks: An Empirical Study. Working paper, SSRN 1476372.

Authors: Sugata Ray

SSRN: papers.ssrn.com/abstract_id=1476372 · DOI

Abstract (from Crossref):

Using a large sample of hedge funds, I study the effects of the high water mark (HWM) on fund performance, risk, and fund closure. I find that as funds fall below the HWM, the standard deviation of future returns increases, the future expected Sharpe ratio decreases, and the incidence of fund closure increases. In addition to supporting predictions from models in the literature, these results resonate well with economic intuition: HWM contracts function as if the fund manager holds call options on the fund’s returns which have varying degrees of moneyness depending on how far the fund is from the HWM.

19. Rossi and Utkus 2021

cites.docx entry 50 · key: rossi2021

Slide 7, Contribution: “Fintech platform design can help (D’Acunto et al. 2019a; **Rossi and Utkus 2021**; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)”

Who Benefits from Robo-advising? Evidence from Machine Learning. Working paper, SSRN 3552671.

Authors: Alberto G. Rossi, Stephen P. Utkus

SSRN: papers.ssrn.com/abstract_id=3552671

Claim in the draft (pp. 3–4, contribution): Cited with D’Acunto et al. (2019a) for the statement that fintech can help households improve their trading performance and asset allocation.

Abstract (September 2020 working-paper draft, from the [//conference.nber.org/conf%5Fpapers/f145218.pdf](https://conference.nber.org/conf%5Fpapers/f145218.pdf) NBER conference copy; the SSRN page could not be opened, so its wording may differ):

We study the effects of a large U.S. hybrid robo-adviser on the portfolios of previously self-directed investors. Across all investors, robo-advising reduces idiosyncratic risk by lowering the holdings of individual stocks and active mutual funds and raising exposure to low-cost indexed mutual funds. It further eliminates investors’ home bias and increases investors’ overall risk-adjusted performance, mainly by lowering investors’ portfolio risk. We use a machine learning algorithm, known as Boosted Regression Trees (BRT), to explain the cross-sectional variation in the effects of advice on portfolio allocations and performance. Finally, we study the determinants of investors’ sign-up and attrition.

Note (inference, not stated by any source found): This paper appears to have been published as Rossi and Utkus, “The diversification and welfare effects of robo-advising,” *Journal of Financial Economics*, 157 (2024), article 103869 (DOI). The evidence is the same two authors and setting, JFE acknowledgments that list the venues where the working paper was presented, and Rossi’s CV listing the JFE paper and no longer the earlier title. Confirm before changing the citation.

Correction flagged in cites.docx: draft says “Albtero”; SSRN working paper; published record: Alberto; apparently now JFE 157 (2024), 103869 under a new title (inference, see entry 50)

20. Barber and Odean 2000

cites.docx entry 5 · key: barber2000_trading

Slide 7, Contribution: “Retail and social trading (**Barber and Odean 2000**; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)”

Trading Is Hazardous to Your Wealth: The Common Stock Investment Performance of Individual Investors. *The Journal of Finance*, 55(2), 773–806.

Authors: Brad M. Barber, Terrance Odean

SSRN: papers.ssrn.com/abstract_id=219228 · DOI

Claim in the draft (p. 4, contribution): Cited with Petajisto (2013) as the literature on differences in performance between professionals and retail traders, which the draft says it offers insight into.

Abstract:

Individual investors who hold common stocks directly pay a tremendous performance penalty for active trading. Of 66,465 households with accounts at a large discount broker during 1991 to 1996, those that trade most earn an annual return of 11.4 percent, while the market returns 17.9 percent. The average household earns an annual return of 16.4 percent, tilts its common stock investment toward high-beta, small, value stocks, and turns over 75 percent of its portfolio annually. Overconfidence can explain high trading levels and the resulting poor performance of individual investors. Our central message is that trading is hazardous to your wealth.

21. Petajisto 2013

cites.docx entry 47 · key: petajisto2018

Slide 7, Contribution: “Retail and social trading (Barber and Odean 2000; **Petajisto 2013**; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)”

Active Share and Mutual Fund Performance. *Financial Analysts Journal*, 69(4), 73–93.

Authors: Antti Petajisto

SSRN: papers.ssrn.com/abstract_id=1685942 · DOI. The SSRN working-paper abstract is a longer first-person version of the published one below.

Claim in the draft (p. 4, contribution): Cited with Barber and Odean (2000) as the literature on differences in performance between professionals and retail traders, which the draft says it offers insight into.

Abstract (published version):

Using Active Share and tracking error, the author sorted all-equity mutual funds into various categories of active management. The most active stock pickers outperformed their benchmark indices even after fees, whereas closet indexers underperformed. These patterns held during the 2008–09 financial crisis and within market-cap styles. Closet indexing has increased in both volatile and bear markets since 2007. Cross-sectional dispersion in stock returns positively predicts performance by stock pickers.

Correction flagged in cites.docx: draft says FAJ vol. 29; published record: FAJ vol. 69

22. Anginer et al. 2024

not in cites.docx · key: anginer2024

Slide 7, Contribution: “Retail and social trading (Barber and Odean 2000; Petajisto 2013; **Anginer et al. 2024**; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)”

Trading Simulations and Real Money Outcomes. *Journal of Behavioral Finance*, 25(4), 436–448 (online 2023).

Authors: Deniz Anginer, Caio Piza, Sugata Ray (University of Alabama), Luqi Xu

[DOI](#)

Abstract (from OpenAlex):

We examine the link between simulation experience and the performance in real money accounts. We find that more active simulator users, simulator users who take on more risk, and simulator users with the best stock-level performance (but not portfolio-level performance) are more likely to open real money accounts. These users are also likely to be more active real money traders and significantly underperform in their real money trades. These results suggest that without proper education regarding the hazards of active trading and accurate performance evaluation, stock simulator users may draw incorrect inferences regarding their trading skill and experience real-money underperformance.

23. Oehler et al. 2016

cites.docx entry 44 · key: Oehler2016Wiki

Slide 7, Contribution: “Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; **Oehler et al. 2016**; Roder and Walter 2019; Oehler and Schneider 2023)”

Benefits From Social Trading? Empirical Evidence for Certificates on Wikifolios. *International Review of Financial Analysis*, 46, 202–210.

Authors: Andreas Oehler, Matthias Horn, Stefan Wendt

SSRN: papers.ssrn.com/abstract_id=2807385 · [DOI](#). The SSRN abstract is a longer version of the published one below.

Claim in the draft (p. 4, contribution): Oehler et al. show traders on social trading platforms do not outperform the market.

Abstract (published version):

Social trading describes the idea that signal providers make their investment decisions available to other investors who follow them. The performance of social trading has hardly been in the focus of research so far. We analyze the performance of 1084 wikifolio certificates used in social trading and issued in 2012 and 2013. We apply factor models to analyze these certificates’ returns and alphas relative to benchmark indexes. We find that, on average, wikifolios do not outperform the market. However, wikifolios with geographical focus provide better performance than those without. Furthermore, the best performing wikifolios earn significant short-term excess returns.

24. Roder and Walter 2019

cites.docx entry 49 · key: RoderWalter

Slide 7, Contribution: “Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; **Roder and Walter 2019**; Oehler and Schneider 2023)”

What Drives Investment Flows Into Social Trading Portfolios? *Journal of Financial Research*, 42(2), 383–411.

Authors: Florian Röder, Andreas Walter

SSRN: papers.ssrn.com/abstract_id=3000929 · DOI

Claims in the draft:

- p. 4, contribution (with Heuer et al. 2017): Investors chase past performance and consider funds in isolation when choosing which portfolios to invest in.
- p. 6, Section 2.1: The certificates are issued by a bank partnered with the social trading platform, which acts as their market maker.
- p. 7, Section 2.1: Röder and Walter obtain data from a similar social trading platform and observe 955 million euros of total flows on their platform from 2012 to 2016.

Abstract (published version):

We investigate investment flows into more than 5,300 social trading portfolios that are issued as structured products and are tradable at a regular exchange. We find that investment flows chase past performance. However, in contrast to mutual fund flows, the flow–performance relation exists nearly exclusively for the best performing social trading portfolios. Flows follow raw returns rather than factor model alphas. Additionally, flows are highly persistent. Finally, social trading portfolios with higher visibility on the web page of the social trading platform as well as traders communicating actively to investors via public comments attract higher inflows.

25. Oehler and Schneider 2023

cites.docx entry 45 · key: oehler2022

Slide 7, Contribution: “Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; **Oehler and Schneider 2023**)”

Social Trading: Do Signal Providers Trigger Gambling? *Review of Managerial Science*, 17(4), 1269–1331. Published online July 2022.

Authors: Andreas Oehler, Julian Schneider

SSRN: papers.ssrn.com/abstract_id=4551715 · DOI

Claims in the draft:

- p. 4, contribution (with Pelster and Breitmayer 2019): Traders on social trading platforms are prone to gambling.
- pp. 4–5, contribution: Named as the closest paper. They find the incentive structure on social trading platforms has very limited downside for the trader and may motivate them to gamble. The draft states the key difference: they identify gambling behavior when a trader is at an extreme end of the performance spectrum relative to other traders, while the draft studies behavior around achieving bonuses.
- p. 31, end of Section 4 (with Eraker and Ready 2015 and Barberis and Huang 2008): Retail investors pursue lottery-type assets despite awareness that expected returns can be negative in the long run.

Abstract (published version):

Social trading—also referred to as copy trading—is an interactive platform-based innovation facilitating visibility and traceability of signal provider trading activities. Based on published portfolio transaction and return track records, platform users can copy one or several signal providers, i.e. delegate their investment decisions, and thereby become signal followers. Allowing signal providers to administer purely virtual portfolios, in combination with a remuneration scheme based on performance fees and high watermarks, creates convex or option-like incentives (Carpenter, *J Finance* 55:2311–2331, 2000; Doering and Jonen, *SSRN J*, 2018). We argue that the incentive structure imposed by social trading providers, including a very limited monetary downside risk for signal providers, may motivate traders to gamble. In this context, we assess the factors that have an impact on signal provider lottery-like stock transactions (Bali et al., *J Financ Econ* 99:427–446, 2011; Kumar, *J Finance* 64:1889–1933, 2009). We provide empirical evidence that signal providers tend to increase the traded relative share of lottery-like stocks when being located at an extreme end of the relative performance spectrum. Furthermore, we provide evidence that underperforming signal providers increase their net exposure towards lottery-like stocks, in turn exposing signal followers to a lottery-like return structure—triggering gambling.

Correction flagged in *cites.docx*: draft says 2022; published record: 2023 for vol. 17(4); 2022 is the online year

26. Callaway and Sant’Anna 2021

cites.docx entry 13 · key: callaway_did

Slide 32, Differences-in-Differences Estimator: “Two-way fixed effects is biased with heterogeneous, staggered effects (**Callaway and Sant’Anna 2021**; Sun and Abraham 2021)”

Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

Authors: Brantly Callaway, Pedro H. C. Sant’Anna

SSRN: papers.ssrn.com/abstract_id=3148250 · DOI · arXiv

Claim in the draft (p. 17, Section 4): One of four papers cited together (with de Chaisemartin and D’Haultfœuille 2020, Goodman-Bacon 2021, and Sun and Abraham 2021) as recent literature pointing out challenges in interpreting the coefficient from two-way fixed effects models when treatment effects are heterogeneous: there is an undesired weighting of the treatment effect when there is heterogeneity across groups. The draft adds that each of these studies proposes alternative methods for variation in treatment timing, but that most require treatment to be absorbing and binary, which does not hold for the bonus structure.

Abstract (SSRN version; the published wording is the same apart from typography):

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DID) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DID setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

27. Sun and Abraham 2021

cites.docx entry 51 · key: sunabraham

Slide 32, Differences-in-Differences Estimator: “Two-way fixed effects is biased with heterogeneous, staggered effects (Callaway and Sant’Anna 2021; **Sun and Abraham 2021**)”

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

Authors: Liyang Sun, Sarah Abraham

SSRN: papers.ssrn.com/abstract_id=3158747 · DOI

Claim in the draft (p. 17, Section 4): One of the four papers cited together on the challenges of interpreting two-way fixed effects coefficients under heterogeneous treatment effects; see entry 13.

Abstract:

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

28. Kosowski et al. 2006

cites.docx entry 38 · key: kosowski2007

Slide 40, Luck or Strategy?: “Luck vs. skill bootstrap (**Kosowski et al. 2006**; Fama and French 2010)”

Can Mutual Fund “Stars” Really Pick Stocks? New Evidence from a Bootstrap Analysis. *The Journal of Finance*, 61(6), 2551–2595.

Authors: Robert Kosowski, Allan Timmermann, Russ Wermers, Hal White

SSRN: papers.ssrn.com/abstract_id=855425 · DOI

Claim in the draft (pp. 22–23, Section 4): Cited with Fama and French (2010) as the luck-versus-skill bootstrap studies from the mutual fund literature that inspired the draft’s bootstrap simulation.

Abstract (published version):

We apply a new bootstrap statistical technique to examine the performance of the U.S. open-end, domestic equity mutual fund industry over the 1975 to 2002 period. A bootstrap approach is necessary because the cross section of mutual fund alphas has a complex nonnormal distribution due to heterogeneous risk-taking by funds as well as nonnormalities in individual fund alpha distributions. Our bootstrap approach uncovers findings that differ from many past studies. Specifically, we find that a sizable minority of managers pick stocks well enough to more than cover their costs. Moreover, the superior alphas of these managers persist.

Correction flagged in cites.docx: draft says “Timmerman”; published record: Timmermann

29. Fama and French 2010

cites.docx entry 25 · key: fama2010luck

Slide 40, Luck or Strategy?: “Luck vs. skill bootstrap (Kosowski et al. 2006; **Fama and French 2010**)”

Luck Versus Skill in the Cross-Section of Mutual Fund Returns. The Journal of Finance, 65(5), 1915–1947.

Authors: Eugene F. Fama, Kenneth R. French

SSRN: papers.ssrn.com/abstract_id=1356021 · [DOI](#)

Claim in the draft (pp. 22–23, Section 4): Cited with Kosowski et al. (2006) as the luck-versus-skill bootstrap studies from the mutual fund literature that inspired the draft’s bootstrap simulation of the 1.6 percentage point bonus-day return jump.

Abstract (published version):

The aggregate portfolio of actively managed U.S. equity mutual funds is close to the market portfolio, but the high costs of active management show up intact as lower returns to investors. Bootstrap simulations suggest that few funds produce benchmark-adjusted expected returns sufficient to cover their costs. If we add back the costs in fund expense ratios, there is evidence of inferior and superior performance (nonzero true α) in the extreme tails of the cross-section of mutual fund α estimates.
